Welcome to Code Crunch!

C5 | CRUNCH LABS: AI & Data Science

[Mon 1-2PM, Wed 12:30-1:30PM] Weekly | Spring 2025

Attendance

Earn a Certificate of Completion by attending at least 70% of the workshops

CODECRUNCH.CS.FIU

Unit #5

Clustering

C5|Crunch Labs: Crunch AI Data Science

Unsupervised Learning

What is Unsupervised Learning?

- model is trained on data that has no labeled outputs.
    - i.e., the data used to train the model does not include predefined categories or target values.

When is an instance Unlabeled?

- an instance is unlabeled if there is no output feature value for that instance

why?

- The main objective is to explore and model the underlying structure of the data
    - such as grouping similar data points or identifying patterns.

Records

Age	Gender	Income	Profession	Tenure	City
35	M	60,000	IT	12	KRK
23	F	90,000	Sales	3	WAW
18	M	12,000	Student	1	KRK
42	F	128,000	Doctor	13	KRK
34	M	63,000	Manager	8	WAW
56	M	82,000	Teacher	30	WAW

Clustering

What is Clustering?

- group similar data points into clusters based on some similarity or distance metric.
  - It is often used for exploratory data analysis, anomaly detection, and pattern recognition.

Cluster Algorithms

- K-Means Clustering
- Hierarchical Clustering
- DBSCAN

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Evaluation

- Evaluating unsupervised learning techniques can be difficult due to the nature of unsupervised learning.
    - unsupervised learning models do not have an absolute truth label

- **evaluation metrics**
    - Silhouette Score
        - Measures how similar points are to their own cluster vs. other clusters.
        - Higher values (close to 1) indicate better clustering.
    - Davies-Bouldin Index (DBI)
        - Ratio of between-cluster variance to within-cluster variance.
        - Higher values indicate better-defined clusters.
    - Calinski-Harabasz Index
        - Measures the average similarity ratio of each cluster with the most similar cluster.
        - Lower values indicate better clustering.

Q&A Session

Attendance

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Join us for the next sessions

Thursdays: 11AM – 12PM & 2PM – 3PM
Friday: 3PM – 4PM
Monday: 12PM – 1PM

RSVP lu.ma/CODECRUNCH

Your feedback helps us improve. Share your thoughts!
Go to our website: go.fiu.edu/codecrunch